

Practical Assignment 3 - Variant and CNV Analysis

Tejas Rao

School of Interwoven Arts and Sciences, Krea University

BIOS331: Microbial Genomics



September 8, 2025

Part I- Data collection

For ease of analysis and interpretability, two separate Galaxy Histories were created, one for variant analysis and one for CNV analysis. Genome sequence data was collected via FasterQ on the India mirror of the Galaxy server. The SRA ID (SRR18675437) for the reference genome wild-type *E. coli* MG1655 was inputted, and the pair-end sequence was downloaded as a zip file locally and then unzipped. The forward and backward reads (in [fastqsanger.gz](https://fastqsanger.org) format) derived from this were uploaded back onto both Galaxy histories. The reads for the Active Site mutant (SRR14920202) were retrieved and uploaded to the variant analysis history. Similarly, the reads for the duplication mutant (SRR14920179) were uploaded to the CNV analysis history.

However, though the reads of the reference was used for De novo assembly (Spades Scaffolds annotated with Prokka + BWA-MEM), as the annotations for NCBI references are better documented, and of a higher quality, it was used for variant calling. The gff, nucleotide fasta and protein fasta of *Escherichia coli* str. K-12 substr. MG1655 (reference: NC_000913) were downloaded from NCBI ([here](#)) and then uploaded on both Galaxy histories.

Job Information	
Galaxy Tool ID	toolshed.g2.bx.psu.edu/repos/luc/ara_tools/fasterq_dump/3.1.1+galaxy1
Job State	ok
Created	Monday Sep 1st 15:11:18 2025 GMT+5:30
Updated	Monday Sep 1st 15:39:31 2025 GMT+5:30
Time To Finish	19 minutes 12 seconds
Command Line	<pre>set -o grep -q pipefail && set -o pipefail; mkdir -p ~/.ncbi && cp ~/data/jwd06/ma...</pre>
Tool Standard Output	<pre>Downloading accession: SRR18675437... spots read : 7,837,428 reads read : 14,874,856...</pre>
Tool Standard Error	empty
Tool Exit Code	0
Job API ID	11ac944876d9bb33ad855783f48f689b9

Job Parameters	
Input Parameter	Value
select input type	accession_number
Accession	SRR18675437
adv	
Define format specification for sequence	@\$ac.\$sn.\$ri
Minimum read length	Not available.
Select how to split the spots	--split-3: write properly paired biological reads into different files and single reads in another file
Dump only biological reads	true

Job Outputs	
Tool Outputs	Dataset
fasterq-dump log	4: fasterq-dump log
Pair-end data (fasterq-dump)	1: Pair-end data (fasterq-dump) a list with 1 fastqsanger.gz pair
	1: Pair-end data (fasterq-dump) a list with 1 fastqsanger.gz pair
Single-end data (fasterq-dump)	2: Single-end data (fasterq-dump) a list with 9 datasets
	2: Single-end data (fasterq-dump) a list with 9 datasets
Other data (fasterq-dump)	3: Other data (fasterq-dump) a list with 9 datasets
	3: Other data (fasterq-dump) a list with 9 datasets

Fig 1: FasterQ Job parameters to Retrieve the reads for the Wild Type via SRA IDs

Part II- Genome Assembly

The pair-end data derived from FasterQ was inputted to Spades. The parameters of the Job are in Fig 2. The quality offset parameter was set to be Illumina. This was determined by looking at the published data on NCBI (Fig 3).

Dataset Information	
Number	15
Name	SPAdes on data 10 and data 9: Assembly graph
Created	Tuesday Sep 2nd 15:16:53 2025 GMT+5:30
Filesize	9.6 MB
Dkey	?
Format	fastq
File contents	contents
History Content API ID	26c75dccc6b16ac8396a61c94f2ee948
History API ID	496d1aa4e95131a (view)
UUID	149487b4-2873-4599-9b73-499519c484d4
Full Path	/data/tnb1/galaxy_db/files/14/9/dataset_149487b4-2873-4599-9b73-499519c484d4.dat
Originally Created From a File Named	assembly_graph.fastq
Tool Parameters	
Input Parameter	Value
Operation mode	Assembly and error correction
Single-end or paired-end short-reads	paired_collection
FASTA/FASTQ file(s): collection	5: Pair-end data (fasterq-dump) a list with 1 fastqganger-gz pair
Type of paired-reads	Default (—pe)
Select orientation of reads	PR (← →)
Use an additional set of short-reads	false
arf	
Nanopore reads	
PacBio CLR reads	
Sanger reads	
Trusted contigs	
Untrusted contigs	
Pipeline options	Nothing selected.
Set coverage cutoff option	false
Select k-mer detection option	auto
Set Phred quality offset	Auto
Select optional output file(s)	Assembly graph Assembly graph with scaffolds Contigs Scaffolds
Job Outputs	
Tool Outputs	Dataset
SPAdes on : Assembly graph	15: SPAdes on data 10 and data 9: Assembly graph
SPAdes on : Assembly graph with scaffolds	16: SPAdes on data 10 and data 9: Assembly graph with scaffolds
SPAdes on : Contigs	17: SPAdes on data 10 and data 9: Contigs
SPAdes on : Scaffolds	18: SPAdes on data 10 and data 9: Scaffolds

Fig 2: SPades Job parameters for De Novo Genome Assembly

Full -

SRX11233467: WGS of laboratory evolved trimethoprim resistant E. coli isolate
1 ILLUMINA (Illumina MiSeq) run: 5.6M spots, 1.7G bases, 491.7Mb downloads

Design: Genomic DNA extracted using standard lab protocols, fragmented to 150 bp and library prepared.

Submitted by: Indian Institute of Science Education and Research

Study: Trimethoprim resistance in E. coli
[PRJNA741586](#) • [SRP325715](#) • [All experiments](#) • [All runs](#)
[show Abstract](#)

Sample: Trimethoprim resistant derivative of E. coli MG1655 after exposure to 300 ng/mL trimethoprim for ~25 generations
[SAMN19881904](#) • [SRS9279282](#) • [All experiments](#) • [All runs](#)
Organism: [Escherichia coli](#)

Library:
Name: TMPR1
Instrument: Illumina MiSeq
Strategy: WGS
Source: GENOMIC
Selection: RANDOM
Layout: PAIRED

Runs: 1 run, 5.6M spots, 1.7G bases, [491.7Mb](#)

Run	# of Spots	# of Bases	Size	Published
SRR14920202	5,591,224	1.7G	491.7Mb	2021-09-23

ID: 15000453

Fig 3: NCBI entry for Active Site Mutant showing that Illumina was used for Sequencing

Part 3: Variant Analysis

This part was conducted in it’s dedicated Galaxy History. Initially, the de novo assembly’s scaffolds and contigs were used for variant analysis of the Active site mutant. This was done by constructing a gff using Prokka, mapping with BWA-MEM, BBTools for variant calling and the SnpEff for annotation. However, due to the need for more variant annotation of the highest possible quality, the NCBI reference’s gff and Nucleotide fasta were used.

As the gff of the NCBI reference was readily available, Prokka was not used. BWA-MEM2 was used to map and align the reads of the mutant. The resulting bam file served as an input for BBTools for Variant calling, considering the NCBI reference.

33: BBTools: call variants on data 20 and data 32

16 lines 19 columns, 56 comments

format vcf database ? size 11.1 KB

Preview

Visualize

Details

Edit

Dataset Information

Number	33
Name	BBTools: call variants on data 20 and data 32
Created	Tuesday Sep 2nd 17:46:31 2025 GMT+5:30
Filesize	11.1 KB
Dbkey	?
Format	vcf
File contents	contents
History Content API ID	26c75dccc616ac8a88fcef924df789e
History API ID	49681aa4e595131a (view)
UUID	49140f8d-a913-4803-8c4e-4701bee40968
Full Path	/data/dnb11/galaxy_db/files/4/9/1/dataset_49140f8d-a913-4803-8c4e-4701bee40968.dat

Tool Parameters

Input Parameter	Value
Bam file	32: SRR14920202.bam
Select reference genome source; a cached reference or one from the history	history
Using reference genome	20: GCF_000005845.2_ASM584v2_genomic.fna
Ploidy	1
Select output format	vcf
Output variant score histogram?	true
Output zygosity histogram?	true
Output quality histogram?	true

Job Outputs

Tool Outputs	Dataset
BBTools: call variants	33: BBTools: call variants on data 20 and data 32
BBTools: call variants on (scorehist)	34: BBTools: call variants on data 20 and data 32 (scorehist)
BBTools: call variants on (zygosityhist)	35: BBTools: call variants on data 20 and data 32 (zygosityhist)
BBTools: call variants on (qualityhist)	36: BBTools: call variants on data 20 and data 32 (qualityhist)

Fig 4: BBTools Job for Variant Calling

SnpEff Build was used to construct a database for variant annotation, while SnpEff eff was used to annotate the vcf output from BBTools. The resulting annotated vcf was converted to a tsv

file using the ‘VCFtoTab-delimited’ tool for easy data wrangling. Selected columns are shown in Table 1. SnpEff eff also outputted an HTML with statistics on the variants, some of which are used to supplement the discussion below.

Sl no	POS	REF	ALT	Kind	Type	Gene
1	49903	T	G	SNP	missense_variant	folA
2	257899	GTGAAGGTAGGTAATGACTCCAACCTATTGATAGTGTGTTTATGTTCA GATAATGCCCGATGACTTTGTCTATGCAGCTCCACCGATTTTGAGAAC GACAGCGACTTCCGTCCCAGCCGTGCCAGGTGCTGCCTCAGATTCA GGTTATGCCGCTCAATTGCTGCGTATATCGCTTGCTGATTACGTGCA GCTTTCCCTTCAGGCGGGATTACATACAGCGGCCAGCATCCGTCATC CATATCACCACGTCAAAGGGTGACAGCAGGCTCATAAGACGCCCA GCGTCGCCATAGTGCGTTCACCGAATACGTGCGCAACAACCGTCTT CCGGAGACTGTCATACGCGTAAAAACAGCCAGCGCTGGCGCGATTTA GCCCGACATAGCCCCACTGTTCTGTCATTCCGCGCAGACGATGA CGTCACTGCCCGGCTGTATGCGCGAGGTTACCGACTGCGGCCTGAG TTTTTAAGTGACGTAAAATCGTGTGAGGCCAACGCCATAATGCG GGCTGTGCGCGCATCCAACGCCATTATGGCCATATCAATGATT TCTGGTGCGTACCGGGTTGAGAAGCGGTGTAAGTGAAGTGCAGTTG CCATGTTTTACGGCAGTGAGAGCAGAGATAGCGCTGATGTCGGCG GTGCTTTTGCGGTTACGCACCACCCCGTCAGTAGCTGAACAGGAGG GACAGCTGATAGAAACAGAAGCCACTGGAGCACCTAAAAACACC ATCATACTAAATCAGTAAGTTGGCAGCATCACC	G	Del (-776)	feature_ablation	insA9 & insB9
3	346287	C	T	SNP	synonymous	yahN
4	803662	C	A	SNP	missense_variant	ybhJ
5	1267605	G	A	SNP	missense_variant	ychA
6	1299494	GCTAAGGAAGGTGCGAACAAGTCCCTGATATGAGATCATGTTTGTC ATCTGGAGCCATAGAACAGGGTTCATCATGAGTCATCAACTTACCTT CGCCGACAGTGAATTCAGCAGTAAGCGCCGTGACACCAGAAAAAGA GATTTTCTTGCCCGATGGAGCAGATTCTGCCATGGCAAAACATGG TGGAAGTCATCGAGCCGTTTACCCCAAGGCTGGTAATGGCCGGCG ACCTTATCCGCTGGAACCATGCTACGCATTCACTGCATGCAGCATT GGTACAACCTGAGCGATGGCGCGATGGAAGATGCTCTGTACGAAAT CGCCTCCATGCGTCTGTTGCCCCGTTATCCCTGGATAGCGCCTTGC CGGACCGCACCAACCATCATGAATTTCCGCCACCTGCTGGAGCAGCA TCAACTGGCCCCGCAATTGTTCAAGACCATCAATCGCTGGCTGGCC GAAGCAGGCGTCATGATGACTCAAGGCACCTTGCTCGATGCCACCA TCATTGAGGCACCCAGCTCGACCAAGAACAAAGAGCAGCAACGCG ATCCGGAGATGCATCAGACCAAGAAAGGCAATCAGTGGCACTTTGG CATGAAGGCCACATTGGTGTGATGCCAAGAGTGGCCTGACCCAC AGCCTGGTCAACCACGCGGCCAACGAGCATGACCTCAATCAGCTGG GTAATCTGCTGCATGGAGAGGAGCAATTTGTCTCAGCCGATGCCGG CTACCAAGGGGCGCCACAGCGGAGGAGCTGGCCGAGGTGGATGT GGACTGGCTGATCGCCGAGCGCCCCGGCAAGGTAAGAACCTTGAA ACAGCATCCACGCAAGAACAACCGCCATCAACATCGAATACATG AAAGCCAGCATCCGGGCCAGGGTGGAGCACCCATTTCGCATCATCA AGCGACAGTTCCGGCTTCGTGAAAGCCAGATACAAGGGGTTGCTGA AAAACGATAACCAACTGGCGATGTTATTCACGCTGGCCAACTGTT TCGGGCGGACCAATGATACGTCAGTGGGAGAGATCTCACTAAAAA CTGGGGATAACGCCTTAAATGGCGAAGAAACGGTCTAAATAGGCTG ATTCAAGGCATTACGGGAGAAAAATCGGCTCAAACATGAAGAA ATGAAATGACTGAGTCAGCCGAGAAGAATTTCCCCGCTTATTCGCA CCTTCC	G	Del (-1)	transcript_ablation	insH2 l
7	1662428	T	C	SNP	missense_variant	ynfF
8	1813315	A	T	SNP	intergenic_region	cedA
9	1905761	G	A	SNP	missense_variant	mntP

10	1937079	C	T	SNP	missense_variant	YebK
11	1978494	ACATAAATGGGTAATGACTCCAACCTATTGATAGTGTTTATGTTTCAG ATAATGCCCCGATGACTTTGTCATGCAGCTCCACCGATTITGAGAACG ACAGCGACTTCCGTCCCAGCCGTGCCAGGTGCTGCCTCAGATTTCAG GTTATGCCGCTCAATTCGCTGCGTATATCGCTTGCTGATTACGTGCAG CTTTCCTTTACGGCGGGATTACATACAGCGGCCAGCCATCCGTCATCC ATATCACCACGTCAAAGGGTGACAGCAGGCTCATAAGACGCCCCAG CGTCGCCATAGTGCCTTACCGAATACGTGCGCAACAACCGTCTTC CGGAGACTGTCATACGCGTAAACAGCCAGCGCTGGCGCGATTAG CCCCGACATAGCCCCACTGTTCGTCCATTTCGCGCAGACGATGAC GTCACTGCCCCGCTGTATGCGCGAGGTACCGACTGCGGCCTGAGT TTTTTAAGTGACGTAAATCGTGTGAGGCCAACGCCCATATGATT GGCTGTTGCCCGCATCCAACGCCATTATGGCCATATCAATGATT TCTGTGCGTACCGGGTTGAGAAGCGGTGTAAGTGAAGTGCAGTTG CCATGTTTTACGGCAGTGAGAGCAGAGATAGCGCTGATGTCCGGCG GTGCTTTTGGCGTTACGCACCACCCCGTCAGTAGCTGAACAGGAGG GACAGCTGATAGAAACAGAAGCCACTGGAGCACCTCAAAAACACC ATCATACACTAAATCAGTAAGTTGGCAGCATCACC	A	Del (-776)	feature_ablation	insA5 & InsB5
12	2173360	ACC	A	Del (-2)	non_coding_transcript	ygff
13	3443670	G	A	SNP	missense_variant	secY
14	3560455	C	CG	Ins (+1)	non_coding_transcript	glpR
15	3938967	C	A	SNP	missense_variant	rbsR
16	4296380	A	ACG	Ins (+2)	intergenic_region	glpP & yjcO

Table 1: Annotated Variant Table

Of all the variants, 10 are SNPs, 4 are deletions and 2 are insertions. Of the SNPs, only one is synonymous. There were 6 transitions and 4 transversions, implying that transitions are 50% more common than transversions in this case, aligning with our expectations. The frequency of amino acid changes are given in Fig 5.

	A	D	E	F	G	I	L	P	S	V
A										1
D			2							
E										
F							1			
G		1							1	
I										
L						1	1			
P							1			
S										
V										

Fig 5: Frequency of Amino Acid changes observed (derived from SnpEff eff):
Rows are reference amino acids and columns are changed amino acids

Coming to the biological relevance of Table 1, we see an SNP in *folA*, a gene that has been documented to confer trimethoprim resistance to this *E. coli* substrain, as it codes for DHFR (trimethoprim's target). As far as the literature goes, there doesn't seem to be any association between the other variants in Table 1 and resistance to trimethoprim.

Part 4: Copy Number Variant Analysis

This analysis was conducted using the CNVkit pipeline on Galaxy. The pair-end data from the duplication mutant and sequence data of the reference was retrieved as described earlier. The forward and reverse reads were mapped using BWA-MEM2. The gff of the reference was converted into bed format for input into CNVkit using the 'GFF-to-BED' converter on Galaxy.

The workflow adopted was according to CNVkit's documentation (Fig 5).

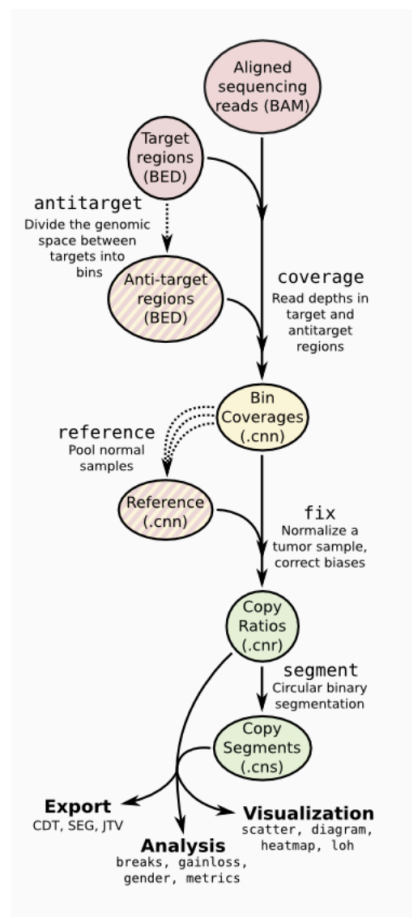


Fig 6: CNVkit Pipeline (Image from <https://cnvkit.readthedocs.io/en/stable/pipeline.html>)

The CNVkit Autobin tool was used to capture the Target and Antitarget regions in the genomic space (The bam file, the reference bed file and nucleotide fasta serve as input). CNVkit Coverage is used to output a cnn file for both CNVkit Autobin outputs of Target and Antitarget. The tool essentially goes through the aligned reads and calculates how much sequencing coverage (how many reads pile up) is present in each of those regions. CNVkit Reference tool was used to calculate a reference important for CNV analysis, using the Autobin outputs and the reference nucleotide fasta. CNVkit Fix takes this as the input, with CNVkit Coverage files, outputting a cnr file (Job parameters given in Fig 7).

30: CNVkit Fix on data 29, data 28, and data 27: Bin-Level log2 Ratios/Coverages

29,755 lines 7 columns
format **cnr** database ? size **4.6 MB**

Preview Visualize Details Edit

Dataset Information

Number	30
Name	CNVkit Fix on data 29, data 28, and data 27: Bin-Level log2 Ratios/Coverages
Created	Tuesday Sep 2nd 23:37:25 2025 GMT+5:30
Filesize	4.6 MB
Dbkey	?
Format	cnr
File contents	contents
History Content API ID	26c75dccc616ac8413d40f8d6a72971
History API ID	e9205efe2cb87fc2 (view)
UUID	b89f7a9d-2a55-40a1-9ffe-47b9a4a99c01
Full Path	/data/dnb11/galaxy_db/files/b/8/9/dataset_b89f7a9d-2a55-40a1-9ffe-47b9a4a99c01.dat
Originally Created From a File Named	sample.cnr

Tool Parameters

Input Parameter	Value
Sample Target coverage cnn file	27: CNVkit Coverage on data 14, data 25, and data 19: Sample Target coverage
Sample Antitarget coverage cnn file	28: CNVkit Coverage on data 14, data 26, and data 19: Sample Target coverage
Reference coverage cnn file	29: CNVkit Reference on data 14, data 26, and data 25: Target on-target coverage
Compare and use cluster-specific values present in the reference profile	false
Sample ID for target/antitarget files	Not available.
Skip GC correction	false
Skip edge-effect correction	false
Skip RepeatMasker correction	false

Job Outputs

Tool Outputs	Dataset
CNVkit Fix on : Bin-Level log2 Ratios/Coverages	30: CNVkit Fix on data 29, data 28, and data 27: Bin-Level log2 Ratios/Coverages

Fig 8: Job Parameters of CNVkit Fix

CNVkit Segment looks at the noisy per-bin ratios from CNVkit Fix and finds contiguous stretches that behave similarly. The output of CNVkit Segment was read into python (pandas), and some of the columns are seen in Fig 8. We see that CNVkit split the genomic space into 103 bins, and assigns them.

```
df.drop(columns=["gene", "chromosome"])
```

✓ 0.0s

	start	end	log2	depth	probes	weight
0	0	3276	-0.155981	542.9640	21	18.19900
1	3276	5304	-0.646175	383.5930	13	11.26600
2	5304	17628	-0.138158	554.4820	79	68.46280
3	17628	83772	0.659765	965.8760	424	367.44600
4	83772	84396	-1.078990	307.4420	4	3.46647
...	...	...	...	...	...	...
98	4476603	4477227	-3.053410	97.4438	4	3.46647
99	4477227	4481283	-0.384508	469.0660	26	22.53210
100	4481283	4539471	0.065273	639.0560	373	323.24900
101	4539471	4543215	-1.312480	287.9500	24	20.79880
102	4543215	4641652	-0.120819	568.2360	631	546.83600

103 rows x 6 columns

Fig 8: CNVkit Segment output as a dataframe

On plotting the average log copy ratio for each segment, we see an increased log ratio initially after which it more or less reduces to the baseline. A single amplified segment is observed on the sole chromosome spanning approximately 1–620,000 bp (≈ 0.62 Mb, $\sim 13\%$ of the 4.6 Mb genome). Now, as the average log ratio for the first 620000 bp is 0.61, we expect that the copy number is $\sim 2 \times 2^{0.61} = 3.053 \approx 3$.

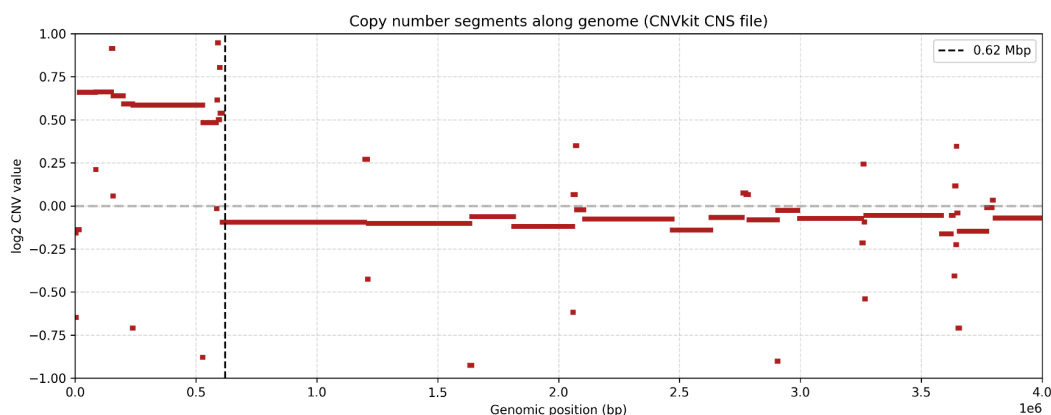


Fig 9: Log Ratio plotted for each CNVkit Segment

An alternate visualisation is in Fig 10, derived from the `cnr` file that was similarly wrangled by pandas. The visualisation recapitulates Fig 9, with elevated $\log_2$ ratio values confined to the same coordinate interval and a background centered near zero elsewhere.

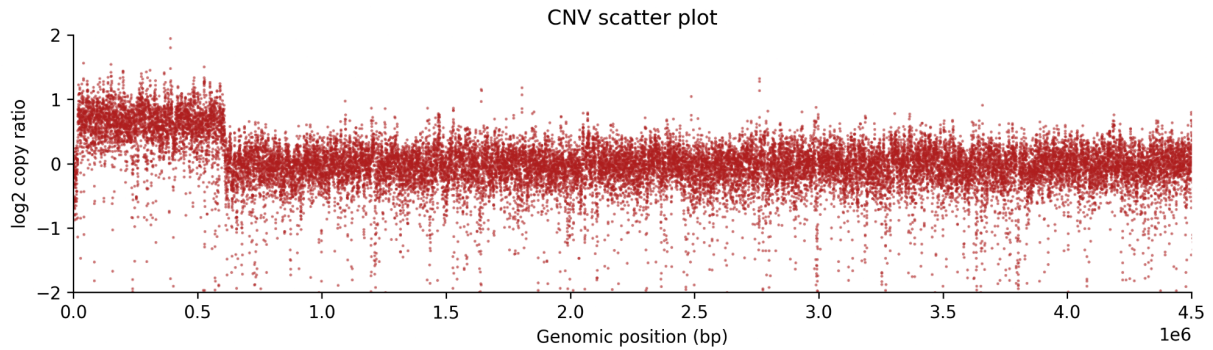


Fig 10: Log Copy Ratio plot from the `cnr` file

Using the NCBI reference `gff`, we see that there are about 523 genes in the amplified region (about 4506 genes present across the entire length of the genome), including the first 3 genes in the variant table (`folA`, `insA9` + `insB9`, `yahN`). The list genes in the the region of length $\sim 620\text{kb}$ is given here:

```
thrL, thrA, thrB, thrC, yaaX, yaaA, yaaJ, talB, mog, satP, yaaW, mbiA, yaaI, dnaK, dnaJ,
insL1, mokC, hokC, sokC, nhaA, nhaR, insB1, insA1, rpsT, yaaY, ribF, ileS, lspA, fkpB,
ispH, rihC, dapB, carA, carB, caiF, caiE, caiD, caiC, caiB, caiA, caiT, fixA, fixB, fixC,
fixX, yaaU, kefF, kefC, folA, apaH, apaG, rsmA, pdxA, surA, lptD, djla, yabP, yabQ, rluA,
rapA, polB, araD, araA, araB, araC, yabI, thiQ, thiP, thiB, sroA, sgrR, sgrS, sgrT, setA,
leuD, leuC, leuB, leuA, leuL, leuO, yabR, ilvI, ilvH, cra, mraZ, rsmH, ftsL, ftsI, ftsO,
murE, murF, mraY, murD, ftsW, murG, murC, ddlB, ftsQ, ftsA, stfZ, ftsZ, lpxC, secM, secA,
mutT, yacG, zapD, coaE, yacM, guaC, hofC, hofB, ppdD, nadC, ampD, ampE, aroP, pdhR, aceE,
aceF, lpd, yacH, acnB, yacL, speD, speE, yacC, cueO, gcd, hpt, can, yadG, yadH, yadI,
yadE, panD, rpnC, panC, panB, yadC, yadK, yadL, yadM, htrE, yadV, yadN, folK, pcnB, gluQ,
dksA, sfsA, thpR, hrpB, mrcB, fhuA, fhuC, fhuD, fhuB, hemL, yadX, clcA, yadW, erpA, yadS,
btuF, mtn, dgt, degP, cdaR, yaeH, yaeI, dapD, glnD, map, tff, rpsB, tsf, pyrH, frr, dxr,
ispZ, ispU, cdsA, rseP, bamA, skp, lpxD, fabZ, lpxA, lpxB, rnhB, dnaE, accA, ldcC, yaeR,
tilS, rof, yaeP, yaeQ, arfB, nlpE, yaeF, proS, trmO, rcsF, metQ, metI, metN, gmhB, rrsH,
ileV, alaV, rrlH, rrfH, aspU, dkgB, yafC, yafD, yafE, mltD, gloB, ykfO, yafS, rnhA, dnaQ,
aspV, yafT, ykfM, yafV, ivy, fadE, gmhA, yafJ, dpaA, yafQ, dinJ, yafL, rayT, dinB, yafN,
yafO, yafP, pepD, gpt, frsA, insB9, insA9, phoE, proB, proA, thrW, ykfI, yafW, ykfH, ykfG,
yafX, ykfF, ykfB, yafY, yafZ, ykfA, perR, insI1, eyeA, ykfC, insH1, mmuP, mmuM, afuC,
insB2, insA2, yagB, yagA, yagE, yagF, yagG, yagH, xynR, argF, argL, ykgS, insB3, insA3,
yagJ, yagK, yagL, yagM, ykgV, yagN, intF, yagP, paoD, paoC, paoB, paoA, yagU, ykgJ, ecpE,
ecpD, ecpC, ecpB, ecpA, matA, ykgL, ykgO, ykgM, ykgR, insE1, insF1, rclC, rclB, rclA,
rclR, ykgE, ykgF, ykgG, ykgH, betA, betB, betI, betT, yahV, pdeL, yahB, yahC, yahD, yahE,
```

yahF, yahG, yahH, yahI, yahJ, yahK, yahL, xtpA, yahM, yahN, yahO, prpR, prpB, prpC, prpD, prpE, codB, codA, cynR, cynT, cynS, cynX, lacA, lacY, lacZ, lacI, mhpR, mhpA, mhpB, mhpC, mhpD, mhpF, mhpE, mhpT, yaiL, frmB, frmA, frmR, yaiO, insC1, insD1, yaiP, ykiC, yaiS, tauA, tauB, tauC, tauD, hemB, ytiB, insF2, insE2, iprA, ampH, sbmA, yaiW, yaiY, yaiZ, ddlA, iraP, phoA, psiF, dgcC, proC, yaiI, aroL, yaiA, aroM, ppnP, ykiD, rdgC, mak, araJ, sbcC, sbcD, phoB, phoR, brnQ, proY, malZ, acpH, queA, tgt, yajC, secD, secF, yajD, tsx, yajI, nrdR, ribD, ribE, nusB, thiL, pgpA, yajO, dxs, ispA, xseB, thiI, yajL, panE, yajQ, yajR, cyoE, cyoD, cyoC, cyoB, cyoA, ampG, yajG, bolA, tig, clpP, clpX, lon, hupB, ppiD, ybaV, fadM, queC, ybaE, cof, decR, mdlA, mdlB, glnK, amtB, tesB, ybaY, ybaZ, ffs, ybaA, pdeB, ylaC, maa, hha, tomB, acrB, acrA, acrR, mscK, ybaM, priC, ybaN, apt, dnaX, ybaB, recR, htpG, adk, hemH, aes, gsk, ybaL, fsr, ushA, ybaK, chiX, ybaP, ybaQ, copA, glsA, ybaT, cueR, ybbJ, qmcA, fetA, fetB, cnoX, ybbO, tesA, ybbA, ybbP, rhsD, ybbC, ylbG, selU, allS, allA, allR, allZ, gcl, hyi, glxR, allW, allB, ybbY, glxK, allE, allC, allD, fdrA, ylbE, ylbF, ybcF, purK, purE, lpxH, ppiB, cysS, ybcI, ybcJ, folD, sfmA, sfmC, sfmD, sfmH, sfmF, fimZ, argU, intD, insE3, insF3, emrE, ylcJ, ybcK, ybcL, ybcM, ylcH, ybcN, ninE, ybcO, rusA, ylcG, quuD, insH2, essD, rrrD, rzpD, rzoD, borD, ybcV, ybcW, ylcI, nohD, tfaD, tfaX, appY, ompT, envY, ybcH, nfrA, nfrB, cusS, cusR, cusC, cusF, cusB, cusA, pheP, ybdG, nfsB, ybdF, ybdJ, ybdK, sokE, hokE, insL2, entD, fepA, fes, ybdZ, entF, fepE, fepC

Both the segmented CNVkit (Fig 9) and the raw bin scatter (Fig 10) show a discrete amplification from 0 to ~0.62 Mb, consistent with a duplication event. The segment mean log₂ ratio of ~0.61 (≈ 0.58) corresponds to an inferred copy number of ~3, indicating that the first 620 kb of the chromosome is present in three copies. No additional broad-scale Copy Number events are apparent across the remaining genome.